Shahyar Anfaz

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Education

University of Toronto — Toronto, ON
Honours B.Sc., Computer Science Specialist and Mathematics Minor — *Sep. 2024 – Expected Apr. 2028*

- GPA: **3.81/4.00**; Dean's List Scholar, 2025 and 2026
- Coursework: Operating Systems, Computer Organization, Systems Programming, Algorithms, Machine Learning

Experience

University of Toronto – AMD-Sponsored Capstone — GitHub
Systems Developer, AMD Schola Godot Port — *Sep. 2026 – Present*

- Built the **C++17 GDExtension** foundation with SCons targets, pinned dependencies, split add-ons, and CI.
- Defined and implemented the episode-lifecycle API for reset, step, terminal/truncation, validation, and step counting.
- Tested **3 auto-reset modes**, terminal-observation preservation, and cross-episode state isolation with Catch2.
- Integrated contributions across a **7-person team**, separated legacy and current demos, and maintained build and release documentation.

University of Toronto — Mississauga, ON
Teaching Assistant, CSC207: Software Design — *Fall 2026*

- Delivered weekly **Java** labs on object-oriented design, unit testing, and design patterns to **150+ students**.
- Guided debugging and refactoring in labs and code reviews, translating defects into actionable feedback.

Projects

skern x86-64 Operating System | *C, C++, x86-64 Assembly, UEFI, QEMU* — GitHub

- Built a custom **UEFI** loader and freestanding **C++** kernel with validated ELF loading and a versioned boot ABI.
- Implemented **GDT/TSS/IDT**, 256 vector stubs, normalized trap frames, and IST-backed critical exceptions.
- Validated boot, fault handling, ABI layout, and register preservation in **QEMU** across **64–512 MiB**.

SHAYVERI Chess Engine v2.10.1 | *C++20, pthreads, AVX2/BMI2, PyTorch* — GitHub

- Reached **3261 STC (10+0.1) / 3321 LTC (90+0.5)** in anchored, Ordo-analyzed multi-engine gauntlets.
- Engineered **Lazy SMP** alpha–beta search, a race-safe shared transposition table, and adaptive time control.
- Implemented incremental **NNUE** with bit-identical scalar/**AVX2** inference and embedded quantized weights.

asdb | *Rust, storage engine, B-tree indexing, TCP/HTTP* — GitHub

- Engineered dependency-free **Rust** storage with **4 KiB pages**, a buffer pool, slotted records, and persistent catalogs.
- Built B-tree indexes plus shared query planning and lazy execution behind **HTTP/JSON** and binary **TCP** APIs.
- Sustained **1.77M inserts/second** in a **10M-key** sequential benchmark on AMD EPYC while passing **239 tests**.

Technical Skills

Languages: C++, C, Rust, Python, Java, SQL, Assembly (x86-64, RISC-V)
Operating Systems: Linux, POSIX, UEFI, bootloaders, interrupts, virtual memory, memory management, IPC
Architecture & Performance: x86-64, multithreading, synchronization, SIMD, AVX2/BMI2, profiling
ML Systems: PyTorch, ROCm, NNUE, quantized inference, reinforcement learning, gRPC, Protocol Buffers
Tools: Git, CMake, Make, Cargo, GCC, GDB, Valgrind, perf, QEMU, ASan, UBSan, CTest, pytest, GitHub Actions